

Intersection of lines

Key Points

To find when two lines r_1 and r_2 intersect set them equal to each other and solve for each coordinate x, y, z checking for consistency

Non-parallel lines in 3D need not intersect and are called skew

Basic Skills

Q1. Given that $r_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $r_2 = \begin{pmatrix} 0 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

- For r_1 write down $x(\lambda), y(\lambda)$
- For r_2 write down $x(\mu), y(\mu)$
- Be equating $x(\lambda) = x(\mu), y(\lambda) = y(\mu)$ find λ and μ
- Hence find the coordinate of intersection
- Find the cartesian equations of r_1 and r_2
- Solve the cartesian intersection to check your answer to d)

Q2. What possibilities are there if we are trying to solve 3 linear equations with only 2 unknowns

Competency Skills

Q3. Given the two lines $r_1 = \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$ and $r_2 = \begin{pmatrix} 7 \\ 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ -1 \\ 3 \end{pmatrix}$

- Show that the point of intersection must satisfy the equation in the x -direction $2 + \lambda = 7 - 2\mu$ and find the other two equations for y and z -directions
- By solving the equations in the x and y -directions find λ, μ
- Check if the z -direction is consistent with your solution of λ, μ and hence state if the lines intersect, stating the point of intersection if so
- Explain how your results would change if the direction of line two was changed to be $-2i - j + 4k$

Q4. Find the point of intersection, if it exists between

a) $r_1 = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ -4 \end{pmatrix}$ and $r_2 = \begin{pmatrix} -1 \\ -3 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ **b)** $l_1: \frac{x-6}{3} = \frac{y-3}{2} = \frac{z}{4}$ and $l_2: \frac{x+2}{2} = \frac{y}{3} = \frac{z+0.4}{10}$

Advanced Skills

Q5. Prove that the lines $r_1 = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 0 \\ 5 \end{pmatrix}$ and $r_2 = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 1 \\ 1 \end{pmatrix}$ are skew

Q6. On a 3D art display the coordinates $(-5, 2, -2)$ is joined by a piece of red string to the point $(5, 3, 2)$ and the centre at $(0, 0, 0)$ is joined by a piece of blue string to the point $(-3, 2, k)$. Assuming the strings are taut

- If $k = -3$ determine if the strings meet
- If $k = 1$ determine if the strings meet
- Determine all k such that strings meet

Extension

Q7. Given that a plane can be written as $r = a + \lambda d_1 + \mu d_2$ determine what sort of lines always have a point of intersection with the plane in 3D

Hint: Compare the number of parameters to the size of the dimension